PHASETRACKER

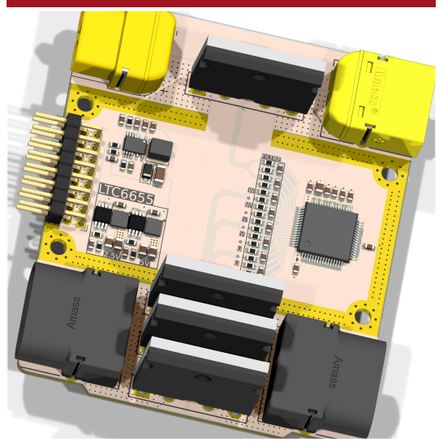
2026-10-19

Variant:

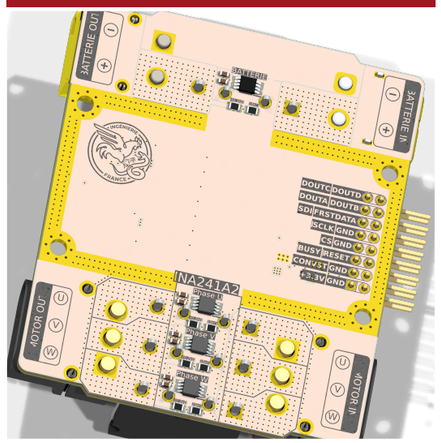
Rev + (Unreleased)

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11	.....	21	.....	31	.....
2	Block Diagram	12	.....	22	.....	32	.....
3	Project Architecture	13	.....	23	.....	33	.....
4	AD7606C-18_ADC	14	.....	24	.....	34	.....
5	INA241_Phase_Sense	15	.....	25	.....	35	.....
6	Power_Supply	16	.....	26	.....	36	.....
7	Connectors	17	.....	27	.....	37	.....
8	Holes, Fiducials	18	.....	28	.....	38	.....
9	Power - Sequencing	19	.....	29	.....	39	.....
10	Revision History	20	.....	30	.....	40	.....

TOP VIEW



BOTTOM VIEW



NOTES

3-phase current and voltage sensing: INA241A2 + AD7606C-18

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 19-Oct-2026

DESIGN CONSIDERATIONS

DESIGN NOTE: Current: 2.5 V - 20 mV/A x I(J1->J2). Voltage: 0.198 V/V.

DESIGN NOTE: No Power-Good: rails within 0.1 % after 71 ms.

DESIGN NOTE: AD7606C digital pins and +3.3V input not wired yet.

DESIGN NOTE: +2.5VA: R12 = 49.9k gives 2.55 V, 24.9k gives 2.50 V. See README.

LAYOUT NOTE: Kelvin-route the shunt sense pads to each INA241.

Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: 04dd92c
	Board Name: PHASETRACKER		Project Name: OPEN_PHASE	
Sheet Title: OPEN_PHASE - PHASETRACKER	File Name: OPEN_PHASE.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-10-06	Revision: + (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 10